MIL-A-48449A(AR)
19 October 1981
SUPERSEDING
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1 February 1974

MILITARY SPECIFICATION

ALIGNMENT DEVICE, OPTICAL, GUN BORE

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of alignment device which provides A close-in optical reference to check the boresight of the fire control at the weapon site. It has a radioactively illuminated reticle.

2.1 APPLICABLE DOCUMENTS

2.1 Government documents-.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2) , the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-O-13830	-	Optical Components for Fire Control Instruments, General Specification Governing the Manufacture, Assembly and Inspection
MIL-F-13926	-	Fire Control Materiel; General Specification Governing the Manufacture and Inspection
MIL-P-14232	-	Parts, Equipment, and Tools for Army Materiel, Packaging, and packing of
MIL-I-45607	-	Inspection of Equipment, Supply and Maintenance of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-810 - Environmental Test Methods

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

F11741648 - Alignment Device, Optical Gun Bore
F8565556 - Pressure Tester, Two Station
F9333945 - Fixture, Final for Align. Device M139/M140

PACKAGING DATA SHEET

11741648 - Packaging of Alignment Device, Optical, Gun Bore

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the test of this specification and the references cited herein, the text of this specification shall take precedence.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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DSAM 4145.8 - Radioactive Commodities in the DOD Supply System.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENTS

3.1 First Article. When specified (see 6.3) , the contractor shall furnish sample unit(s) for First Article Inspection and approval (see 4.2 and 6.3).

3.2 Regulations pertaining to radioactive devices. Marking, identification, handling, and shipping of radioactive devices shall be in accordance with DSAM 4145.8.

3.3 Materials. Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.4 Fabrication. The alignment device shall be manufactured in accordance with Drawing F11741648 and drawings pertaining thereto.

3.5 General specifications. The following requirements of MIL-F-13926 shall be met:

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Parts identification and marking
e. Workmanship

3.6 Orientation. Whenever specified herein, the conditions listed below shall apply:

a. Dovetail base surface in a horizontal plane.
b. Side of dovetail parallel to a 4 power optical reference telescope.
c. Alignment device latched against side of dovetail.

3.7 Temperature conditions. Except as specified in 3.8.1, 3.8.2, 3.9.6; the requirements of this specification shall be met at a standard ambient temperature between 60° and 90°F (16° to 32°C).

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3.8 Environmental service conditions.

3.8.1 Storage temperature. The alignment device shall be capable of withstanding temperature stabilization at +160°F (71°C) and -50°F (-46°C) for a period of 4 hours at each temperature extreme. Subsequent to return to standard ambient temperature, the alignment device shall not have been damaged or adversely affected and shall meet the requirements of 3.9 (see 4.6.2).

3.8.2 Operating temperature. Unless otherwise specified, the alignment device shall meet the requirements of 3.9.6.2 while thermally stabilized at temperatures of +150°F (66°C) and -40°F (-40°C). Subsequent to return to room temperature, the alignment device shall not have been damaged or adversely affected and shall meet the requirements of 3.9 (see 4.6.3).

3.8.3 Shock. The alignment device in its Carrying Case (or a simulated case) shall withstand a total of 18 shock impulses, 3 in each direction of 3 mutually perpendicular axis. Each shock impulse shall be a half-sine wave with a time duration of 11 + 1 milliseconds. The peak amplitude of each shock impulse shall–be 75g's. Subsequent to shock, the alignment device shall not have been damaged or adversely affected and shall meet the require-ments of 3.8.5 and 3.9 (see 4.6.4).

3.8.4 Vibration.

3.8.4.1 Vibration "A". The alignment device in its Carrying Case (or a simulated case) shall withstand vibration for 15 minutes each in 3 mutually perpendicular directions. Vibrations in each direction shall be at frequencies varied logarithmically from 5 to 500 Hz. The double amplitude of applied vibrations shall be .036 inches at lower frequencies and shall be maintained at, but not exceed, a 5g level at higher frequencies. Subsequent to vibration, the alignment device shall not have been damaged or adversely affected and shall meet the requirements of 3.8.5 and 3.9 (see 4.6.5.1).

3.8.4.2 Vibration "B". With the alignment device held by the upright tube in a horizontal position, the alignment device shall-be-vibrated in a vertical plane at a constant frequency of thirty Hz with an amplitude of 1/16 inch (1/8 inch total excursion) for a period of five minutes plus or minus 15 seconds. Subsequent to vibration, the Alignment Device shall not have been damaged or adversely affected and shall meet the requirements of 3.8.5 and 3.9 (see 4.6.5.2).

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3.8.5 Effects of vibration and shock. This requirement shall be met with conditions of 3.6 reestablished. The line of sight established prior to vibration or shock shall not deviate more than .1 mil in azimuth due to subjecting the Alignment Device to Vibration "A" , Vibration “B”, or Shock (see 4.6.6).

3.9 Performance.

3.9.1 Sealing. The interior of the alignment device shall be capable of maintaining 10 pounds per square inch gage (psig) for a period of 2 hours + 10 minutes at a temperature of 750 + 15°F. At the end of the 2 hour period, the leakage shall not have exceeded 0.2 psig. The pressure within the alignment device shall then be returned to 1 to 2 pounds per square inch gage (psig) and the optical cavity sealed. The interior atmosphere of the optical cavity of the accepted alignment device assembly shall be nitrogen in accordance with 3.9.2 (see 4.6.7).

3.9.2 Purging. The interior of the alignment device shall be flushed with dry nitrogen until the gaseous mixture bled from the alignment device has a dew point of at least as low as -25°F (see 4.6.8).

3.9.3 Cleanliness.

3.9.3.1 Dirt. This requirement shall be met with the observations for dirt and foreign particles (classed as dirt) made through the objective end of the alignment device. The maximum dimension of any particle appearing on the reticle surface shall not exceed the width of the reticle line. There shall be not more than three particles visible on the reticle. There shall be no dirt in the optical system which shall interfere with observation or adversely affect image quality. particles smaller than 1/4 reticle line width shall be ignored regardless of distribution (see 4.6.9).

3.9.3.2 Defects. There shall be no evidence of moisture, finger prints, condensates, fractures, and adhesive separations on any glass component. Internal inspection shall be made by viewing through the objective end of the alignment device (see 4.6.9).

3.9.4 Accuracy.

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3.9.4.1 Collimation. With conditions of 3.6, the line of sight of the device reticle shall be parallel in elevation with the base of the dovetail within 0.1 mil and parallel in azimuth within 0.1 mil to the side (angular surface) of the dovetail (see 4.6.10).

3.9.4.2 Replacement repeatability. When the alignment device is removed from the reference dovetail mount and subsequently remounted, the device reticle line of sight shall meet the requirements of 3.9.4.1 and shall not have changed from the original position by more than 0.1 mil (see 4.6.10).

3.9.5 Optical characteristics.

3.9.5.1 Parallax. Using a collimator with the reticle adjusted to be parallax free for a 80 meter target, the parallax between the reticle and the cross on the alignment device reticle shall not exceed 0.1 mil (see 4.6.11).

3.9.6 Operability of moving parts.

3.9.6.1 Attachment. The alignment device shall meet all functional requirements of this specification when the alignment device mounting bracket or a suitable substitute is employed (see 4.6.10).

3.9.6.2 Latch. The latch shall be free to rotate to lock the alignment device to the dovetail on the alignment device's mounting bracket and shall be free to rotate to release the boresight from the dovetail. This requirement shall be met at standard ambient conditions and at the temperatures specified in 3.8.2 (see 4.6.10).

3.9.7 Illumination.

3.9.7.1 Reticle. The reticle marking shall be clearly distinguishable when observed in ambient light conditions ranging from dusk into night through a 4-power telescope at a distance of 10 feet $\pm$ 1 foot. This observation must be made with and without the use of a 50% neutral density filter (see 4.6.12).

3.9.8 External tritium contamination. Any part of tritium shall not be present on the external surfaces of the alignment device in excess of 1000 dpm/100 sq. cm. subsequent to the requirements of 3.8.1 through 3.8.4.2 (see 4.6.13).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, -the contractor may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to-perform- any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. The component and subassembly inspection requirement of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval. The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) alignment devices plus three (3) each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production (see 4.4.2) .

4.2.1 Government testing. When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing. When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor ‘s inspection.

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4.3 Inspection provisions.

4 .3.1 Submission of product. Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with “Submission of Product” provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies. All components and subassemblies shall be inspected in accordance with the applicable Quality Assurance Provisions of MIL-F-13926.

4.3.2.2 Final acceptance inspection. Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II, and 111 herein. Examination and tests for packaging, packing, and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in Tables I, II, and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I. 100% Inspection

No.	Characteristic	Requirement	Test procedure
101.	Orientation	3.6	4.6.1
102.	Vibration “B”	3.8.4.2	4.6.5.2
103.	Effects of Vibration & Shock	3.8.5	4.6.6
104.	Sealing	3.9.1	4.6.7
105.	Purging	3.9.2	4.6.8
106.	Cleanliness	3.9.3	4.6.9
107.	Reticle accuracy (Collimation)	3.9.4.1	4.6.10
108.	Repeatability	3.9.4.2	4.6.10
109.	Parallax	3.9.5.1	4.6.11
110.	Operability	3.9.6	4.6.10
111.	Reticle Illumination	3.9.7.1	4.6.12
112.	Radiological Contamination	3.9.8	4.6.13

NOTE : The tests in Table I shall be conducted at a temperature between +60 and +90°F.

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4.3.3 Acceptance and rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with “Acceptance and Rejection” as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General. Three alignment devices shall be selected at random by a Government representative as a special sample from each 50 produced. The samples shall meet the requirements and tests in Table II and shall then meet the requirements and tests in Table I.

TABLE 11. Special sampling

No.	Characteristics	Requirements	Test procedures
201.	Fabrication	3.4	Applicable drawings/ Visual
202.	General specification	3.5	MIL-F-13926/Visual
203.	Storage temperature	3.8.1	4.6.2
204.	Operating temperature	3.8.2	4.6.3

4.4.2 First article testing. Inspection shall be performed on the following characteristics during First Article Testing (See 4.2). Prior to performing these tests, the selected items shall have met the inspection criteria outlined in Tables I and II. Upon successfully meeting the inspection criteria outlined in Table III, the selected items shall be reinspected for those characteristics of Table I.

TABLE III. First article testing

No l	Characteristics	Requirements	Test procedures
401.	Shock	3.8.3	4.6.4
402.	Vibration “A”	3.8.4.1	4.6.5.1

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4.4.3 Failure of sample. Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (See 4.3.3).

4.5 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply and maintain equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment. Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the Technical Data Package (TDP).

4.5.2.2 Contractor design. The contractor shall design and supply inspection equipment comparable with the “Test Methods and Procedures” specified in 4.6 of this specification and with the component inspection procedures specified in “Examination” and “Test Facilities” requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of the test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Test equipment. In conjunction with 4.5.2.2, the following Standard Test Equipment shall be utilized in the performance of the applicable tests specified in 4.6.

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Nomenclature	Description
1. Temperature Chamber; Vibration Equipment; and Shock Equipment.	Standard type conforming to the requirements of MIL-STD-810 and capable of providing the specified test conditions.

4.6 Test methods and procedures.

4.6.1 Orientation. Test fixture conforming to Drawing F9333945 will be used for the inspection of the Alignment Device, Optical, Gun Bore. Affix-the test fixture on a leveled, vibration free surface in accordance with the specified set-up instructions. The calibration and set-up instructions are the prime requirements necessary to establish the conditions of 3.6, prior to positioning the Alignment Device under test to the test fixture. The Alignment Device under test shall be oriented as specified in 3.6. Operating instructions for final inspection fixture are detailed on Drawing F9333945.

4.6.2 Storage temperature. The Alignment Device shall be exposed to each temperature extreme specified in 3.8.1 for a minimum of 4 hours. After each temperature exposure, the alignment device shall be stabilized at standard ambient conditions and subjected to the examinations and tests to determine conformance with the requirements of 3.8.1. Caution shall be exercised during this test to avoid subjecting the alignment device to thermal shock.

4.6.3 Extreme operating temperature. The Alignment Device shall be exposed to each temperature extreme specified in 3.8.2 for a minimum of 4 hours. After exposure for 4 hours and while at the test temperature, operability of latch shall be in accordance with 3.9.6.2. After-each temperature exposure, the Alignment Device shall be stabilized at standard ambient conditions for a minimum of 4 hours and subjected to the examinations and tests to determine conformance with the requirements of 3.8.2. Caution shall be exercised during this test to avoid subjecting the alignment device to thermal shock.

4.6.4 Shock. The alignment device in its carrying case or equivalent shall be tested in accordance with Method 516 of MIL-STD-81O, Procedure I, except as specified in 3.8.3. Subsequent to testing, the unit shall be subjected to the examinations and tests to determine conformance with the requirements of 3.8.3.

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4.6.5 Vibration.

4.6.5.1 Vibration "A". The Alignment Device while in its carrying case (or equivalent), shall be vibrated in accordance with-Procedure IX, Part 1 of MIL-STD-81O, except as specified in 3.8.4.1. Upon completion of the vibration cycling, the alignment device shall be subjected to the examinations and tests to determine conformance with the requirements of 3.8.4.1.

4.6.5.2 Vibration "B". The alignment device shall be secured to the vibration table by suitable adapters with the tube and head parallel to the vibration table surface. Lever , 10544452, shall be immobilized. The alignment device shall then be vibrated in accordance with the conditions specified in 3.8.4.2. Upon completion, the unit shall be subjected to the examinations and tests to determine conformance with the requirements of 3.8.4.2.

4.6.6 Effects of Vibration and Shock. Subsequent to vibration or shock, the alignment device shall be thoroughly examined for physical defects resulting from poor workmanship and materials to determine conformance with 3.8.5, prior to being subjected to the remaining tests specified herein. A visual examination shall be made for missing or damaged parts. A tactile examination shall also be conducted to determine looseness or improper functioning of moving parts. An audible examination shall be made of the unit by shaking it and listening for loose internal parts or broken optical components, such as lenses and reticle, etc. Conditions of 4.6.1 shall be reestablished and the line of sight established prior to vibration or shock shall not have deviated in excess of the tolerance specified in 3.8.5. Any one of the possible defects defined above shall be cause to consider an alignment device defective. Items which have passed the above examinations shall be subjected to the tests specified in 4.6.7 through 4.6.13.

4.6.7 Sealing. The internal pressurizing of the alignment device to test sealing shall be Performed with special testing equipment conforming to Drawing F8565556, Pressure Tester-Two Station. The pressure shall be applied through an appropriate adapter. Pressurize the unit with dry nitrogen gas to 10.0 $\pm$ 0.10 psig. The dry nitrogen gas shall have a dew point at least as low as minus 25°F. The applied pressure shall be retained for a minimum of 2 hours. Any evidence of leakage at the end of the 2 hour time period, shall not exceed the tolerance specified in 3.9.1, as measured on the pressure gage of the testing equipment.

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4. 6.8 Purging. Upon satisfactory completion of the sealing test, the alignment device shall be flushed with dry nitrogen gas. The flushing shall be accomplished by opening the purging vent and allowing the nitrogen gas to flow through the Alignment Device until the bled gas conforms to the dew point requirements of 3.9.2 as measured with a dew point tester. Close the purging vent.

4.6.9 Cleanliness. The Alignment Device, with Collimator Assembly D7573433, shall be subjected to the applicable test procedures outlined in MIL-O-1383C to determine compliance with the requirements of 3.9.3.1 and 3.9.3.2.

4.6.10 Reticle accuracy (collimation) . With the Alignment Device positioned as specified in 4.6.1, observe where the Alignment Device's boresight cross falls upon the target in the Collimator Assembly. The intersection of the boresight cross shall not deviate from the collimator target zone in excess of the limit specified in 3.9.4.1. Remove and reposition the unit under test three times. At each resetting into the fixture, the boresight cross in the alignment device shall remain centered within the specified limits to ascertain compliance with requirement of 3.9.4.2.

4.6.11 Parallax. Test shall be performed using the set-up in 4.6.1 with the 80 meter focus in the Collimator Assembly. Parallax shall be recognized as any apparent displacement, when the observer's head is moved from side to side or up and down. Any movement between the collimator's reticle and the alignment device reticle shall not exceed the value specified in 3.9.5.1. (The width of a reticle line in the collimator is equal to 0.10 mil).

4.6.12 Reticle Illumination. The test shall be performed in an area in which ambient lighting can be varied. Personnel performing the testing are to be dark adapted by being in the darkened room for at least 15 minutes before viewing the reticle. The reticle shall be viewed through the collimator assembly using a 4-power telescope at a distance of 10 feet $\pm$ 1 foot. The reticle marking shall be clearly distinguishable with and without the use of a 50% neutral density filter while the light level is varied gradually from O to 10 foot candles over a period of 1 to 2 minutes.

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4.6.13 Radiological contamination. Contamination test shall be performed by wiping the Alignment Device with filter paper moistened in distilled water. The wiping shall be performed with moderate finger pressure. The damp filter paper shall be placed in a container with the proper portion of scintillation liquid. The scintillating system used to measure contamination must be calibrated. The readout shall be within 10% of the known standard value when counted to a total of 2000 disintegrations. The actual test for determining contamination is performed by placing the container into the scintillating system. The measured contamination shall be within the limits specified in 3.9.8.

5. PREPARATION FOR DELIVERY

5.1 packaging, packing, and marking. Packaging and packing shall be in accordance with Packaging Data Sheet 11741648. Marking shall be in accordance with MIL-STD-129. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. The alignment device is used to provide an on-carriage optical reference to periodically check the boresight setting of the fire control.

6.2 Ordering data . Procurement should specify the following:

a. Title, number, and date of this specification.
b. Selection of applicable levels of preservation, packing, and packaging.
c* Applicable stock number.
d. Provisions of First Article testing.

6.3 First Article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation. The first article sample shall consist of the first production items and as specified in 4.2. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examinations, test, and approval of the first article.

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6.4 Radioactive material. Prospective bidders should be cognizant of the need for an Atomic Energy Commission license governing the procurement and handling of-radioactive material.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-A-48449A Alignment Device, Optical, Gun Bore

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